Girls Build Bots — Junior Robotics Badge Curriculum

A badge-aligned workshop guide for robotics outreach events

Level: Juniors (Grades 4–5) · **Badges covered:** Programming Robots, Designing Robots, Showcasing Robots **Format:** One full-day camp (6 hours) or three 2-hour sessions · **Ratio:** 1 mentor per 4–5 Scouts

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Robotics badge series. Official badge requirements are available in your Girl Scout Volunteer Toolkit (VTK). Please verify completed activities against the official steps before awarding badges. Badges are purchased by troops through your council shop.

Big Ideas for Juniors

Juniors are ready for **real programming logic and structured design**. They should leave able to:

1. Write a program with **sequences, loops, and events** — not just single commands.
2. Explain a robot's **parts** (sensors, controller/brain, body, motors, tools) and how they work together.
3. Use the **engineering design process**: define the problem → brainstorm → plan → build → test → improve.

Vocabulary to introduce: loop, event, condition, input/output, sensor, actuator, iteration, prototype, engineering design process.

Module 1: Programming Robots (2 hours)

Badge theme: Creating programs a robot could run, using real programming structures.

Materials

- Laptops with **MakeCode Arcade** (arcade.makecode.com)
- "Unplugged loops" cards (dance-move cards with repeat counters)
- Projector for live coding
- Optional hardware: micro:bit boards or any drivable robot the team owns (great but not required)

Activity 1.1 — Loop Dance Warm-Up (15 min)

Post a dance routine as cards: *clap ×2, spin ×1, jump ×3.* Perform it. Then rewrite it with a "repeat" card: *[clap, spin] repeat 2.* **Debrief:** loops let programmers avoid writing the same steps over and over.

Activity 1.2 — Build a Chase Game in MakeCode Arcade (60 min)

Mentor-led progressive build, one concept per stage: 1. **Sprites & events** — create a player, move with arrow keys (*event: on button pressed*) 2. **A second sprite** — add a "battery" the player must collect (*event: on overlap*) 3. **Loops** — make the battery respawn in a random spot *forever* 4. **Score & condition** — add score; **if** score reaches 5, show "YOU WIN" 5. **Challenge round** — add an enemy sprite that follows the player

Scouts who finish early become "deputy mentors" for a neighbor — this doubles as leadership practice and keeps the room moving.

Activity 1.3 — Program Like a Roboticist (30 min)

Connect the game to robots: the player sprite = robot, arrow keys = sensor inputs, the win condition = the robot's goal. Scouts then write **pseudocode on paper** for a real robot task (e.g., a vacuum robot: *forever: move forward; if bump sensor pressed, turn right*). Share two or three aloud.

If hardware is available, run one pseudocode plan on a real robot as a group demo.

Completion Checklist — Module 1

- ☐ Can explain sequences, loops, and events with an example
- ☐ Built a working game with events, a loop, and a win condition
- ☐ Wrote pseudocode for a robot behavior
- ☐ Debugged at least one problem in her own program

Module 2: Designing Robots (2 hours)

Badge theme: Learning how robots are built and designing/building a robot model that solves a problem.

Materials

- Robot parts diagram poster (sensors, controller, chassis, motors, effectors)
- Build materials: cardboard, motors-optional craft supplies, LEGO/K'NEX if available, hot glue station (mentor-operated)
- Engineering design process posters and planning sheets (problem, sketch, materials list, test plan)

Activity 2.1 — Robot Anatomy Teardown (20 min)

Using photos (or a real robot), identify the five part groups: **sensors** (eyes/ears), **controller** (brain), **chassis** (body), **motors/actuators** (muscles), **effectors** (hands/tools). Quick game: mentor names a robot ("Mars rover"), Scouts call out one part and its job.

Activity 2.2 — Design Sprint: Robots That Help (70 min)

Teams of 2–3 follow the engineering design process end to end: 1. **Define (10 min):** pick a "helping" problem from a challenge menu — helping an elderly neighbor carry groceries, finding a lost pet, watering a garden, cleaning up a park. 2. **Brainstorm & plan (15 min):** sketch the robot; label its sensors, brain, movement, and tool; list materials. 3. **Build (30 min):** construct the model. It should *represent* every labeled part (a bottle cap can be a camera). 4. **Test & improve (15 min):** teams present to a mentor who plays "skeptical customer" and asks one hard question; teams make one improvement.

Activity 2.3 — Design Review Circle (30 min)

Each team gives a 90-second design pitch: the problem, the parts, the one improvement they made. Peers give "two stars and a wish" feedback (two things they liked, one suggestion).

Completion Checklist — Module 2

- ☐ Can name a robot's main part groups and their jobs
- ☐ Followed all steps of the engineering design process
- ☐ Built a labeled robot model that addresses a real problem
- ☐ Made at least one improvement based on testing/feedback

Module 3: Showcasing Robots (2 hours)

Badge theme: Sharing robot designs with an audience and connecting to the wider world of robotics.

Materials

- Team models and design sheets from Module 2
- Poster boards or slide template (one simple 4-slide frame)
- Optional: FRC robot demo, short video of a FIRST competition match

Activity 3.1 — The World of Robotics (25 min)

High school team presents: what FIRST robotics is, what competitions look like, and the many roles on a team (programmer, builder, strategist, outreach). Show a 2-minute match video. Emphasize: **there is a place in robotics for every interest.**

Activity 3.2 — Build Your Showcase (45 min)

Teams turn their design into a presentation using a 4-part frame: 1. The problem we chose and why it matters 2. Our robot: parts and how it works 3. What went wrong and how we fixed it 4. What we'd build next

Format is the team's choice: poster, slides, or live "demo theater" with the model. Mentors coach delivery: every member speaks, practice twice.

Activity 3.3 — Showcase Fair (40 min)

Run it like a mini science fair: teams at stations, audience (other Scouts, leaders, parents) rotates through. Each visitor asks at least one question. Mentors award fun superlatives — "Most Creative Sensor," "Best Debugging Story" — so every team is recognized.

Wrap-Up (10 min)

Reflection round: "What surprised you about building a robot?" Troop leaders confirm badge completion against the VTK.

Completion Checklist — Module 3

- ☐ Learned about robotics competitions and team roles
- ☐ Created a presentation covering problem, design, and iteration
- ☐ Presented to an audience and answered questions
- ☐ Reflected on what she would build next

Mentor Guide (High School Robotics Students)

- **Ask, don't tell.** Juniors can reason: "What do you think the robot needs to know before it turns?"
- **Let teams struggle for 3 minutes** before stepping in — productive struggle is where learning happens.
- **Rotate roles inside teams** so no Scout only watches: builder, recorder, tester, presenter.
- **Debug out loud.** When you help fix code, narrate your thinking so Scouts hear the process.
- **Keep the challenge menu real.** Juniors engage hardest with problems about people they know (grandparents, pets, neighbors).

Pre/Post Survey (Impact Measurement)

1–5 scale, before and after: 1. I know how to write a program with a loop. 2. I can name the main parts of a robot. 3. I feel confident presenting my ideas to a group. 4. I can see myself doing robotics or engineering someday.

Report the average change per question as your outcome metric.

Created for the "Girls Build Bots" Girl Scout Gold Award project. This is an original curriculum aligned to the Junior Robotics badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.